

Programming best practices: Fundamentals of Object Orientation

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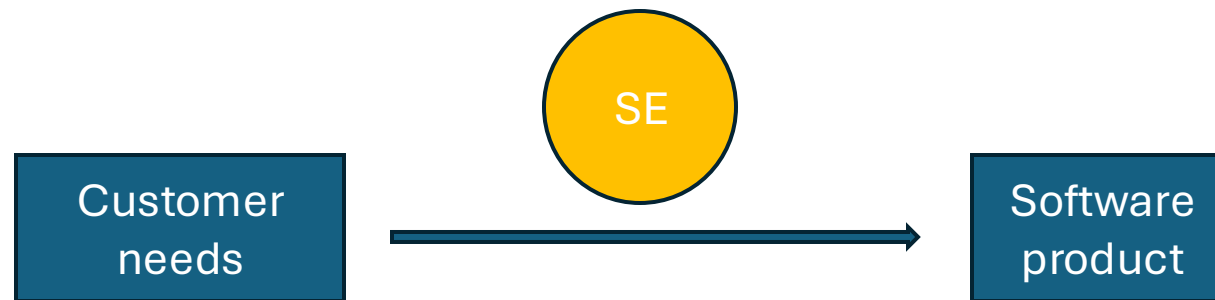
RESCUER Project

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1. Introduction to object orientation

- The goal of Software Engineering (SE) is to find ways to build software with quality
 - "SE = Coordinated application of techniques, methodologies and tools to produce high-quality software, within a given budget and by a given date"



Software quality criteria

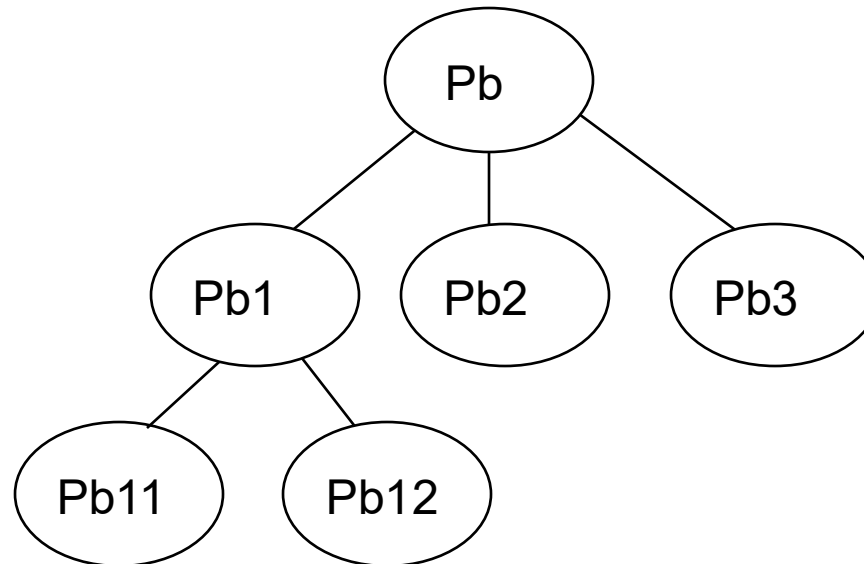
- Attributes that define software quality:
 - Functionality: The degree of certainty that the system processes the information accurately and completely.
 - Flexibility: The degree to which the user may introduce extensions or modifications to the information system without changing the software itself.
 - Maintainability: The ease of adapting the information system to new demands from the user, to changing external environments, or in order to correct defects.
 - Reusability: The degree to which parts of the information system, or the design, can be reused for the development of other applications.
 - Other: efficiency, user-friendliness, portability ...



Modularity



- Modularity (divide and conquer) is one of the most direct mechanisms to achieve the objectives of flexibility, maintainability, and reusability:



Object Orientation

- Methodology: collection of methods, notations, processes and tools applied throughout the development process (analysis, design, implementation, testing, ...)
- Object-Oriented methodologies:
 - They use an object model as an abstraction of the system to be developed
 - The system is seen as a collection of interacting objects
 - An object is a concept, abstraction or physical entity with well-defined boundaries and meaning in the problem domain
 - Example: *Donald Trump*, *Google*, or a *table* are objects
 - Relationships are established between objects that reflect the structure of the real world

Object model

- The object model facilitates the modularity
 - Objects are small modules that encapsulate state and behaviour
 - The state represents the current values of the properties (attributes and bindings) of the object.
 - The state of the object evolves over time
 - The state of the object recalls the effect of the operations on itself
 - Behavior is the set of operations of the object
 - Behavior explains the object's competences: the actions and reactions it can perform

Gradual transition between analysis, design and implementation

- During analysis (what does it do?) and design (how is it solved?) UML is commonly used
 - UML = Unified Modeling Language
 - Widely used notation for object-oriented modelling
 - Includes graphical notations for constructing diagrams to represent specific aspects of the model
 - The static structure of the system by means of class diagrams, package diagrams, deployment diagrams, ...
 - The behaviour of the system through use case diagrams, interaction (sequence) diagrams, ...
- Object-oriented programming languages are used for the implementation, which directly support the concepts of the object model

Key concepts inherent in the object model

- Abstraction: focusing on the essentials
- Encapsulation: hiding implementation details
- Hierarchy: ordering or classification of abstractions
- Polymorphism: sending syntactically the same messages to objects of different types
- Modularity: division into cohesive and loosely coupled modules

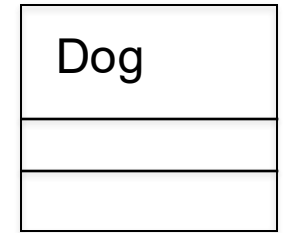
2. Object Oriented Programming in Python

- We will introduce the key concepts of object-orientation and its implementation in Python
 - It reduces the complexity of programs and improves its maintainability
 - As opposed to primitive data structures (lists, tuples, dictionaries), users can define their own data structures to naturally model the reality of the problem they want to solve
- The examples integrated in the following slides are available in this GitHub repository:
 - <https://github.com/javierni/oop-examples>

2.1. Abstraction

- It allows you to focus on the inherent essentials of an entity, and ignore its incidental properties
- The main task in the abstraction process is the identification of objects
 - The object model should reflect the problem domain.
- A class describes a group of objects that share properties (attributes and operations), constraints, relationships with others and have common semantics
 - Person, Company or Table are classes
- Representation of classes in UML:
name, attributes, operations

Class name
attributes
operations



Defining classes in Python

- Definition of a class

```
class <Class name>:
```

```
# dog1.py

class Dog:
    pass
#pass allows error-free code execution
```

Code

- Instantiation: creation of objects

```
<Class name>()
```

```
>>> a = Dog()
>>> b = Dog()
>>> a == b
False
```

Python Terminal

Attributes

- *name, age* and *weight* are attributes of *Person*
 - Describe the properties of the objects
- Each attribute has a value in each instance
- Instantiate is the operation of creating objects of a class
 - Instance = object belonging to a class
- When attributes have the same value in all instances, they are called *class attributes*
- The name of the attribute is unique within the class
- You can specify the type of the values of an attribute

Person
name: string age: integer weight: float

Operations and methods

- An operation is a transformation that can be applied to/by the objects of a class
 - *hire, fire, pay_dividends* can be operations of a *Company* class
 - *sing, laugh* and *jump* can be operations of a *Person* class
- Objects of a class share the same operations
- Operations have arguments
 - The number of arguments, their type and the return type of an operation is called *signature*
- A method is the implementation of an operation for a class
 - Constructors: methods that implement instantiation operations

Person
name age weight address
changeAddress

GeometricObject
colour position
move(delta:Vector) select(p:Point): boolean

Definition of class attributes and instance attributes in Python

Dog
<u>species = "Canis familiaris"</u>
name
age

- Instance attributes are defined within the `__init__()` constructor
 - Can have any number of parameters
 - First parameter: reference to the instance just created (Python sends it automatically during instantiation)
 - By convention it is usually called *self* but another identifier could be used
- Class attributes are defined outside the `__init__` method and an initial value must be assigned
- Notation `.<attribute name>` for accessing the attribute value

```
# dog2.py
class Dog:
    # Indentation to indicate that an
    attribute/method belongs to the class
    species = "Canis familiaris"
    def __init__(self, name, age):
        self.name = name
        self.age = age
```

```
>>> miles = Dog("Miles", 4)
>>> buddy = Dog("Buddy", 9)
>>> miles.name
'Miles'
>>> buddy.name
'Buddy'
>>> buddy.species
'Canis familiaris'
>>> Dog.species = "Perro"
>>> buddy.species
'Perro'
```

Definition of instance methods in Python

Dog
<u>species = "Canis familiaris"</u> name age
description speak(sound)

- Methods defined within a class using indentation
 - The first parameter is the *self* reference (no need to pass it in the invocation)
- Notation *.<method name>* for accessing the method of an object

```
# dog3.py
class Dog:
    species = "Canis familiaris"
    def __init__(self, name, age):
        self.name = name
        self.age = age
    def description(self):
        return f"{self.name} is {self.age} years old"
    def speak(self, sound):
        return f"{self.name} says {sound}"
```

```
>>> miles = Dog("Miles", 4)
>>> miles.description()
'Miles is 4 years old'
>>> miles.speak("Woof Woof")
'Miles says Woof Woof'
```


2.2. Encapsulation

- It's the process of hiding all details of an object that do not contribute to its essential characteristics
- Hiding information about the implementation of an object means that no part of a complex system depends on the internal details of particular objects
- Encapsulation and abstraction go hand in hand:
 - abstraction focuses on the external view
 - and encapsulation on the internal view

Levels of visibility

- UML defines 3 levels of visibility for the members of a class:
 - Private (for the implementers of the class):
 - A private attribute can only be accessed by the class where it is defined
 - A private operation can only be invoked by the class where it is defined
 - Private attributes and operations are not accessible by subclasses or any other class
 - Protected (for extension/subclass creators):
 - A protected attribute or operation can be accessed by the class where it is defined and any descendant of the class
 - Public (for users of the class):
 - A public attribute or operation can be accessed by any class

Tournament
- maxNumPlayers: int # protectedAttribute:int + publicAttribute: int
+ getMaxNumPlayers():int + getPlayers(): List + acceptPlayer(p:Player) + removePlayer(p:Player) + isPlayerAccepted(p:Player):boolean

Encapsulation in Python

- There are no specific modifiers to distinguish between *private*, *protected* and *public*
- By convention, non-public members use an identifier starting with an underscore (`_`), but this does not prevent direct access

Dog

- name
- age

+ set_age(new_age:int)
+ get_age():int
+ set_name(new_name:string)
+ get_name():string

```
# dog4.py
class Dog:
    def __init__(self, name, age):
        self._name = name # Private attribute
        self.set_age(age)

    def set_age(self, new_age):
        self._age = new_age # Private attribute

    def get_age(self):
        return self._age
```

```
>>> miles = Dog("Miles",4)
>>> miles._age = 6 # Not recommended
>>> miles._age # Not recommended
6
>>> miles.set_age(7) # Recommended
>>> miles.get_age() # Recommended
7
```

Name mangling

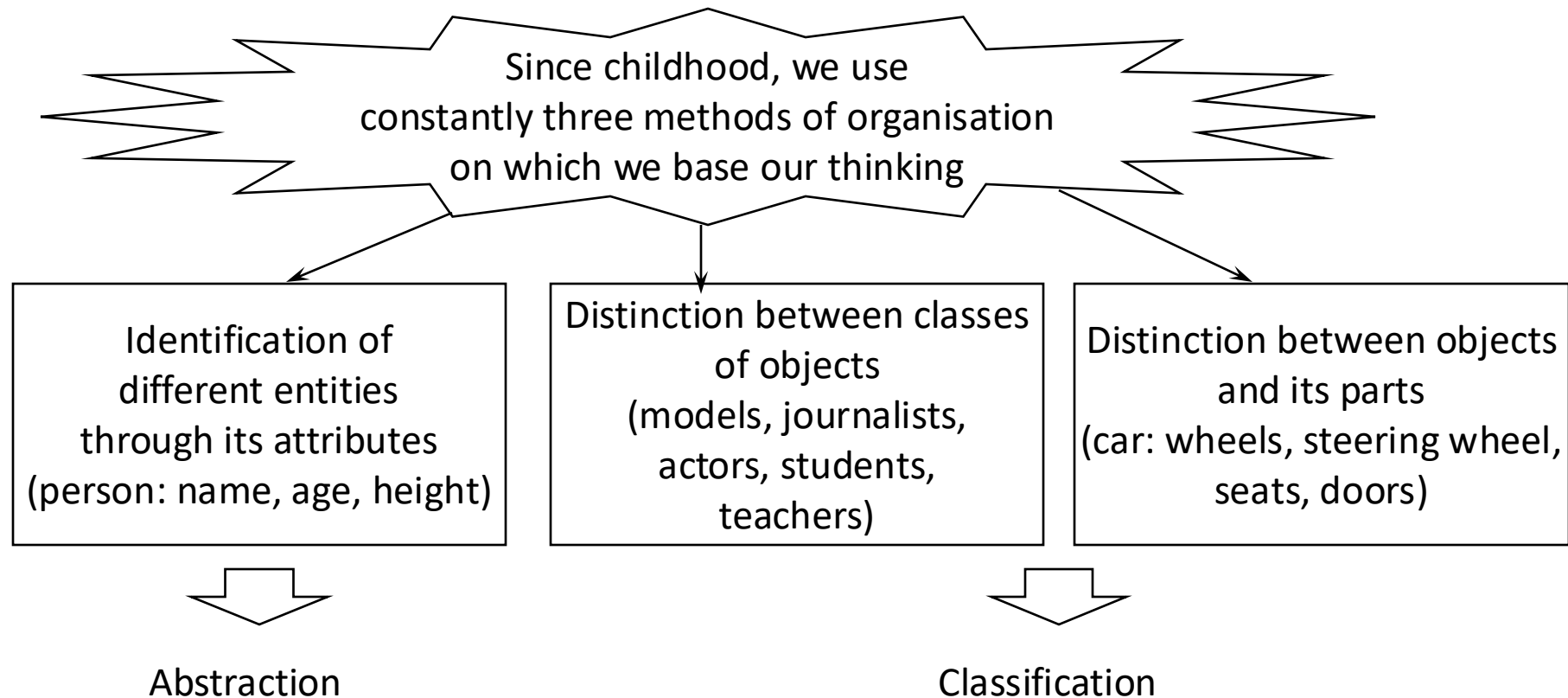
- To reinforce the encapsulation, a double underscore (__) can be used as a prefix
 - This causes automatic renaming (name mangling) and the identifier becomes `__<Class name>__<member name>`
 - Direct accesses with `__<member name>` cause an error

```
class Dog:
    def __init__(self, name, age):
        self.set_name(name)
        self.set_age(age)
    def set_age(self, new_age):
        self.__age = new_age # Private attribute
    def get_age(self):
        return self.__age
    def set_name(self, new_name):
        self.__name = new_name # Private attribute
    def get_name(self):
        return self.__name
```

```
>>> miles = Dog("Miles",4)
>>> miles.__age
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
AttributeError: 'Dog' object has no
attribute '__age'
>>> miles.__Dog__age
4
```

2.3. Hierarchy

- It's the ordering or classification of abstractions

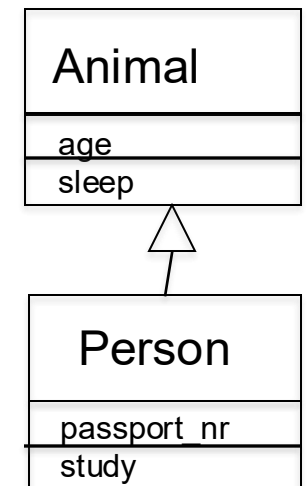
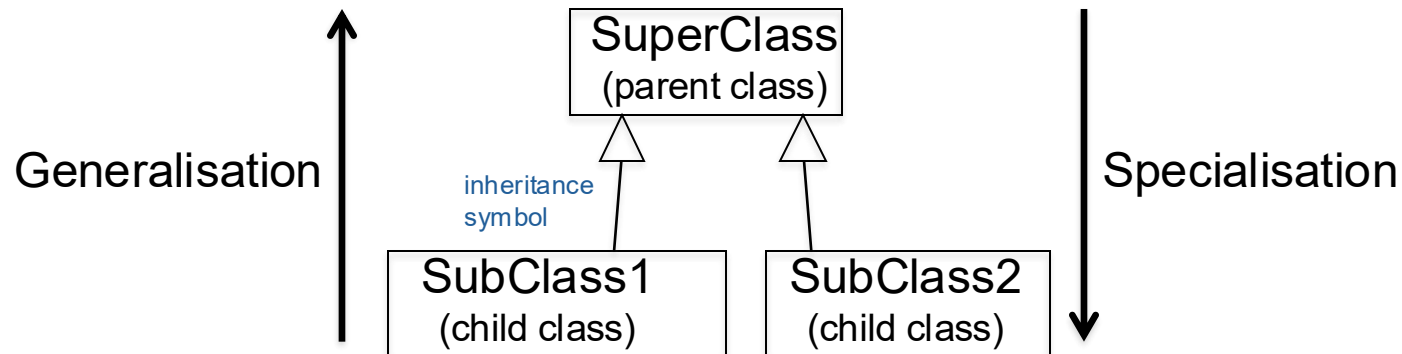


Types of hierarchies

- When we model a system we realise that very few classes are isolated
 - Most of them collaborate with others in various ways
- Main types of relationships/hierarchies between objects
 - Inheritance
 - It's a generalisation/specialisation relationship
 - Association/aggregation/composition
 - It's a structural relationship

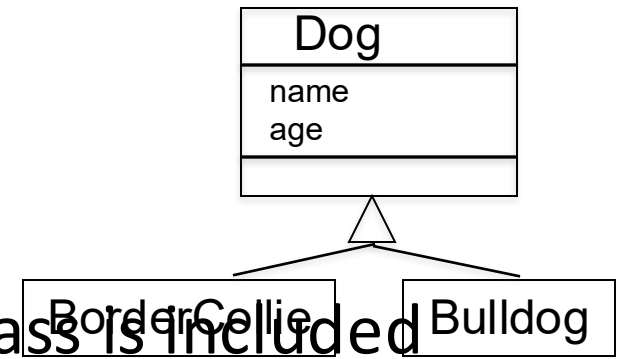
Inheritance

- It's an abstraction to share similarities between classes while maintaining their differences
- It's also known as the “is-a” relationship
- Each instance of a subclass is an instance of the superclass
- Subclasses inherit attributes and methods from the superclass and can add their own
 - Inherited or derived part
 - Emergent or incremental part



Inheritance in Python

- When defining the subclass, the name of the parent class is included in parentheses



```
# dog6.py
class Dog:
    def __init__(self, name, age):
        self.name = name
        self.age = age
    # __str__: special method returning a string representation
    def __str__(self):
        return f"{self.name} is {self.age} years old"

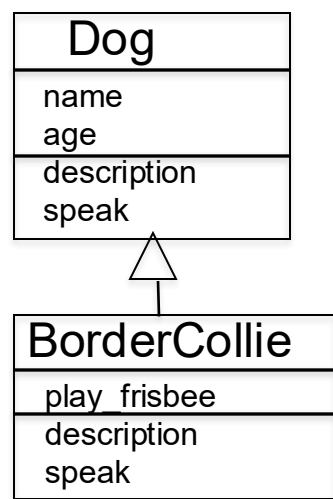
class BorderCollie(Dog):
    pass

class Bulldog(Dog):
    pass
```

```
>>> robert = BorderCollie("Robert", 8)
>>> print(robert)
Robert is 8 years old
>>> type(robert)
<class '__main__.BorderCollie'>
>>> isinstance(robert, BorderCollie)
True
```


New members and redefinition of methods in subclasses

- Apart from redefining the behavior, it's possible to reuse the behavior of the parent class with the reserved word *super*



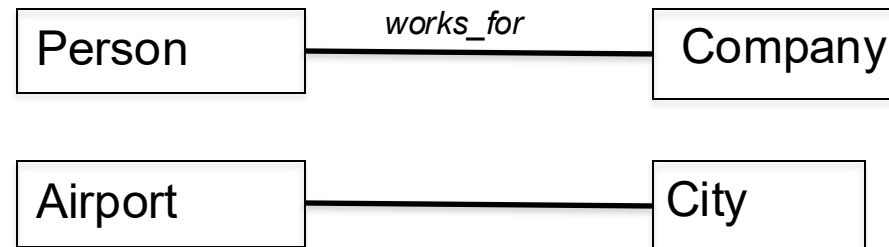
```
# dog7.py
class Dog:
    def __init__(self, name, age):
        self.name = name
        self.age = age
    def description(self):
        return f"{self.name} is {self.age} years old"
    def speak(self, sound):
        return f"{self.name} says {sound}"
```

```
>>> robert = BorderCollie("Robert", 8, True)
>>> robert.description()
'Robert is 8 years old and plays frisbee'
>>> robert.speak() 'Robert says Wow'
```

```
class BorderCollie(Dog):
    def __init__(self, name, age, playfrisbee):
        super().__init__(name, age)
        self.playfrisbee = playfrisbee
    def description(self):
        if self.playfrisbee:
            return f"{self.name} is {self.age} years old and plays frisbee"
        else:
            return f"{self.name} is {self.age} years old"
    def speak(self, sound="Wow"):
        return super().speak(sound)
```

Association

- An association is a relationship that indicates that there is a physical or conceptual connection between objects
- Basic concepts
 - Name of the association: the association can have a name that describes the nature of the relationship
 - Roles (see below)
 - Multiplicity (see below)



Roles

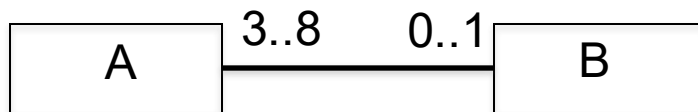
- The role that a class plays in the association can be explicitly stated
- The role is the face that one class presents to the class at the other end of the association



- A class can play the same or different roles in other associations
- Roles can be seen as an alternative to the name of the association
- Roles can coexist with the name of the association

Multiplicity

- It indicates how many objects of one class relate to a single instance of the other class in the association.
- In short: Given an object of class A, how many objects of class B does it relate to at least and at most?
- Representation: *MinValue .. MaxValue*
- Examples:

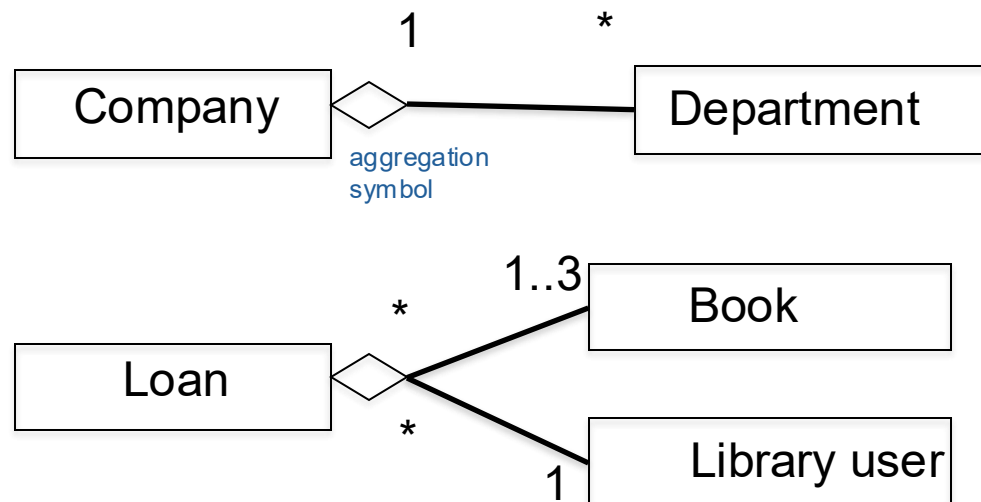


- If nothing is stated, the multiplicity is 1..1
- 1 is equivalent to 1..1
- * is equivalent to 0..*



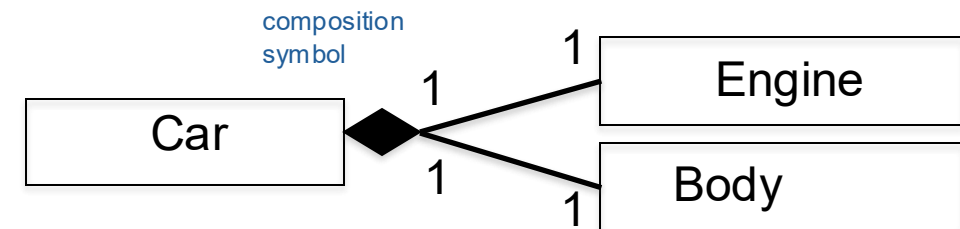
Aggregation

- An aggregation is a special form of association
- It models a “whole/part” or “has a” relationship
- A class (the “whole”) consists of smaller elements (the “parts”)



Composition

- A composition is a more restrictive, stronger form of aggregation
- An aggregation does not link the lives of the “whole” and the “parts”
- An aggregation only distinguishes the “whole” from the “parts”
- Composition:
 - It’s a relationship of belonging and coinciding lives of the “part” with the “whole”
 - The “parts” can be created after the “whole”, but once created they live and die with the “whole”
 - The “whole” manages the creation and destruction of the “parts”
 - The “parts” have no independent existence



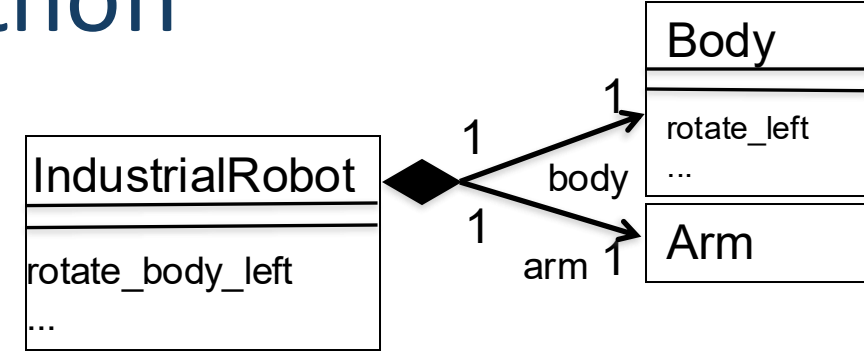
Structural relationships in Python

- The most common relationships are binary relationships (between 2 classes)
- The roles at the extremes of the association/composition/aggregation are translated into instance attributes
- If the maximum multiplicity is 1, these attributes are initialised with an instance of the related class
- If the maximum multiplicity is *, we need to manage the objects associated with a linear data structure (list, set, ...)

Example of 1:1 composition in Python

```
# robot.py
class IndustrialRobot:
    def __init__(self):
        self.body = Body()
        self.arm = Arm()

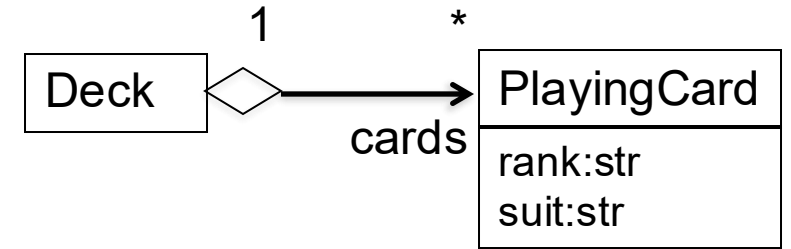
    def rotate_body_left(self, degrees=10):
        self.body.rotate_left(degrees)
    def rotate_body_right(self, degrees=10):
        self.body.rotate_right(degrees)
    def move_arm_up(self, distance=10):
        self.arm.move_up(distance)
    def move_arm_down(self, distance=10):
        self.arm.move_down(distance)
    def weld(self):
        self.arm.weld()
```



```
>>> robot = IndustrialRobot()
>>> robot.rotate_body_left()
Rotating body 10 degrees to the left...
```

```
class Body:
    def __init__(self):
        self.rotation = 0
    def rotate_left(self, degrees=10):
        self.rotation -= degrees
        print(f"Rotating body {degrees} degrees to the left...")
    def rotate_right(self, degrees=10):
        self.rotation += degrees
        print(f"Rotating body {degrees} degrees to the right...")
```


Example of 1:* aggregation in Python



- If the main purpose of the class is to store data, data classes can be used (from Python 3.7 onwards)
 - Simplified definition of instance attributes (suggested data types), string representation, ...

```
#cards.py
from dataclasses import dataclass
from typing import List
```

```
@dataclass
```

```
class PlayingCard:
```

```
    rank: str
```

```
    suit: str
```

```
@dataclass
```

```
class Deck:
```

```
    cards: List[PlayingCard]
```

```
    def add_card(self, card):
```

```
        self.cards.append(card)
```

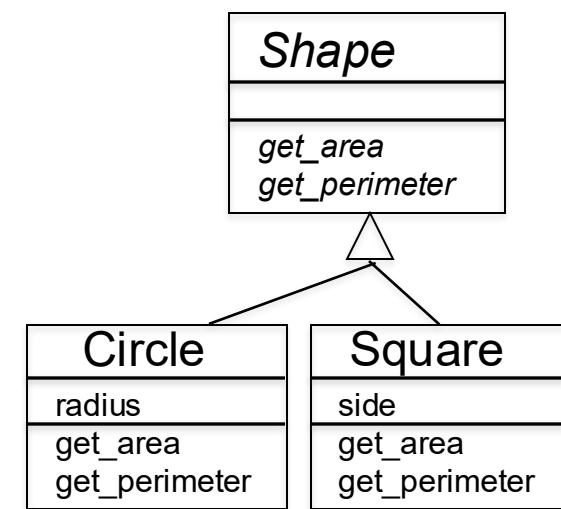
```
>>> queen_of_hearts = PlayingCard('Q', 'Hearts')
>>> ace_of_spades = PlayingCard('A', 'Spades')
>>> deck = Deck([queen_of_hearts, ace_of_spades])
>>> king_of_diamonds = PlayingCard('K', 'Diamonds')
>>> deck.add_card(king_of_diamonds)
>>> deck
Deck(cards=[PlayingCard(rank='Q', suit='Hearts'),
PlayingCard(rank='A', suit='Spades'),
PlayingCard(rank='K', suit='Diamonds')])
```

2.4. Polymorphism

- It's the property by which it is possible to send syntactically equal messages to objects of different types
- Even if the message is the same, different objects can respond to it in unique and specific ways
- Combined with inheritance, it allows for more flexible and adaptable code
 - Objects of derived classes can be treated in the same way as if they were direct instances of the base class

Example with a hierarchy of figures

- *get_area* message sent to an object of unknown type
- The correct *get_area* method is executed via dynamic linking



```
# shapes.py
from abc import ABC, abstractmethod
from math import pi
# abstract class: no direct instances
class Shape(ABC):
    @abstractmethod
    def get_area(self):
        pass # abstract method to be redefined in subclass
    @abstractmethod
    def get_perimeter(self):
        pass
    def __str__(self):
        return f"object with area {self.get_area()} and perimeter {self.get_perimeter()}"
```

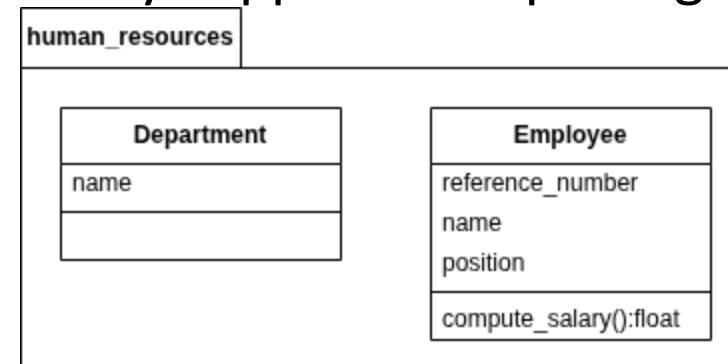
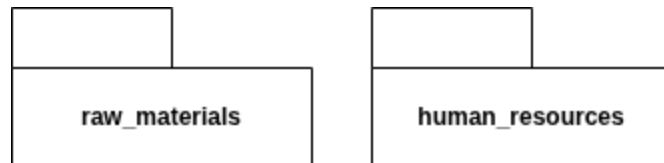
```
class Circle(Shape):
    def __init__(self, radius):
        self.radius = radius
    def get_area(self):
        return pi * self.radius ** 2
    def get_perimeter(self):
        return 2 * pi * self.radius
```

```
>>> shape_a = Circle(100)
>>> print(shape_a) object with area 31415.926535897932 and perimeter 628.3185307179587
```

2.5. Modularity

- It's the property of a system that consists of a set of “cohesive” and loosely “coupled” modules
- Apart from considering classes (objects) as small modules, packages are defined to logically group the classes of a system
 - Cohesive module: it groups together closely related classes
 - Loosely-coupled module: it attempts to minimise the dependencies between a module and the other modules of a system.
 - Package diagrams are included in UML
 - Object-oriented programming languages usually support the “package” concept

package symbol



Module management in Python

- A module is a grouping of related code (classes if we use object orientation) stored in a single .py file.
- A package is a collection of related modules
 - It's physically stored in a directory structure
- A library is a collection of related packages and modules
- The recommendations seen in the session on "Modularisation and code style" also apply here

3. Exercises

- [Click to open the exercises](#)
- Ex1: Definition of attributes and methods
 - Construction of a class to represent a time of day
- Ex2: Encapsulation
 - Definition of operations in a class to represent fractions
- Ex3: Inheritance
 - Define a class to represent squares in a hierarchy of figures
- Ex4: Aggregations
 - Definition of operations between polynomials expressions

4. References

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- The Python Tutorial. <https://docs.python.org/3.10/tutorial/index.html>